

LESSON 14.2

DETERMINING SAMPLE SPACE USING OUTCOME TABLES



LESSON INTRODUCTION

MA.8.DP.2.1 Determine the sample space for a repeated experiment.

LEARNING TARGETS

- I can determine the sample space for a repeated experiment using lists, tables, and tree diagrams.
- I can determine the sample space for a two-step experiment.

BENCHMARK COHERENCE

BUILDING ON

MA.7.DP.2.1 Determine the sample space for a simple experiment.

WORKING TOWARD

MA.912.DP.4.1 Describe events as subsets of a sample space using characteristics, or categories, of the outcomes; or as unions, intersections or complements of other events.

ESSENTIAL QUESTIONS

- How can sample spaces be determined and represented?
- How can the number of outcomes be determined for the sample space of a repeated experiment?

LESSON NARRATIVE

Students will build on the skill of determining the sample space of random processes by learning to represent the sample space in outcome tables. Then, students develop a method for determining the size of the sample space. The primary learning outcome in this lesson focuses on identifying the sample space of an experiment using various methods as well as developing strategies for calculating the number of outcomes in a repeated experiment.

COMPONENT SUMMARY

14.2.1 WARM-UP (~5 MIN.)

Students determine the sample space for a repeated experiment using a list

14.2.3 GUIDED PRACTICE (10 MIN.)

Determine the number of outcomes in a sample space

14.2.5 YOUR TURN! (5-10 MIN.)

Students practice determining sample space and sample space size

14.2.2 COLLABORATION (10 MIN.)

Introduction to using outcome tables to determine sample space

14.2.4 BRING IT TOGETHER (5-10 MIN.)

Make connections to develop a method for determining sample space size

14.2.6 WRAP-UP (~5 MIN)

Students demonstrate their ability to determine sample space and sample space size

RESOURCES

GLOSSARY

Key Terms:

- Outcome table

Additional Terms:

- N/A

MATERIALS

- (Optional) Graph paper
- (Optional) Rulers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may list sample spaces without putting them in set notation.
- Students may misinterpret the outcomes in the tables as not being unique, i.e., the outcome (2,5) differs from (5,2) on a die.

THINGS TO CONSIDER

- Students need to have the correct answers for question 6 in the 14.2.2 Collaboration, question 2 in the 14.2.3 Guided Practice, and question 1 in the 14.2.4 Bring It Together in order to notice the pattern for determining the sample space. Monitor for student errors in these specific questions before 14.2.4 Bring It Together.

LITERACY AND LANGUAGE ROUTINES

Poll the Class

In 14.2.2 Collaboration, consider using this language routine to determine students' benefits and weaknesses of using tree diagrams versus outcome tables to determine sample space. Facilitate a class discussion on the strengths and weaknesses of both representations for students to complete the table together.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 14.2.4 Bring It Together, students use the repeated experiments explored throughout this lesson to look for similarities among the questions. Positioning students to use patterns and structure to help understand and connect different models of sample space size will help the students to develop generalizations that can be applied to other questions. Understanding these relationships will also help students better understand how to determine the theoretical probability of repeated experiments in upcoming lessons.

DIFFERENTIATE INSTRUCTION

This activity provides a multitude of opportunities for differentiation throughout. Multiple representations provide opportunities for students to demonstrate their understanding in a variety of ways. 14.2.3 Guided Practice encourages students to choose the best representation for determining sample space, allowing student choice to utilize the method that makes the most sense to them. Encourage students to think flexibly about the different approaches for determining sample space and consider scenarios in which each method makes sense throughout this unit.

Additional differentiation opportunities are denoted throughout the Lesson Guide.

Answers will vary.

	Heads	Tails
1	H1	T1
2	H2	T2
3	H3	T3
4	H4	T4
5	H5	T5
6	H6	T6

14.2.1 WARM-UP: BELL WORK

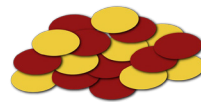
TEACHER MOVES

If time allows, consider pairing students to discuss their answers to see if peers can help work through anything they are struggling with and to check each other's work.



14.2.1 Warm-Up: Bell Work

- The integer chips shown each have a yellow side to represent a positive integer and a red side to represent a negative integer.



Complete the table for the random process below.

Number of Flips of an Integer Chip	Sample Space
1	$\{R, Y\}$
2	$\{RR, RY, YR, YY\}$
3	$\{RRR, RRY, RYR, YRR, RYY, YRY, YYR, YYY\}$



14.2.2 Collaboration: Outcome Tables

Another way to organize the outcomes in a sample space is with an outcome table.



Outcome tables are a useful way to list and organize all possible outcomes in a random process with two steps. They are especially useful when there are a large number of outcomes in the sample space.

One limitation of an outcome table is that it cannot be used to determine the sample space of experiments with more than two steps.

Below is an outcome table showing the possible outcomes when a coin is flipped two times.

		Toss 2	
		Heads	Tails
Toss 1	Heads	HH	HT
	Tails	TH	TT

Since there are six possible outcomes when one standard six-sided die is rolled, the outcome table for rolling two six-sided dice needs to be six by six.

In the outcome table shown, the numbers 1-6 in the second column from the left represent the possible outcomes when the first die is rolled.

- Complete the grey row to represent the possible outcomes from rolling the second die.

		Die 2					
		1	2	3	4	5	6
Die 1	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

The pairs of numbers in the first white row of the outcome table represent all outcomes that can occur when a 1 is rolled on the first die. The outcome (1, 3) means the first die rolled is a 1, and the second die rolled is a 3.

- Complete the next row of the outcome table to show the possible outcomes when a 2 is rolled on the first die.
- What patterns do you see in the outcome table at this time?
Answers will vary. All of the first numbers are a 2. The numbers from the second row count in order from 1 to 6.
- Work with your partner to complete the rest of the outcome table.

14.2.2 COLLABORATION: OUTCOME TABLES

TEACHER MOVES

MA.K12.MTR.2.1: Multiple Representations

After giving students time to read this page, consider pausing the class to ensure understanding and provide the opportunity to make connections between different representations. Help students make connections between lessons through questioning: In the previous lesson, we created a list and tree diagram for flipping a coin two times. How does this outcome table compare to that list and tree diagram? Which representation do you prefer? Why?

TEACHER MOVES

Although students should notice the patterns, they may follow the pattern of the table and mindlessly fill in the last rows without thinking of what each outcome represents. Bring attention to some of the outcomes in the final rows and have students explain their meaning.

For example:

- What does the outcome (6, 5) represent?
- Where is the outcome of rolling a 4 on the first die and a 3 on the second?
- How does the outcome (2, 5) compare to (5, 2)?

14.2.2 COLLABORATION: OUTCOME TABLES (CONTINUED)

TEACHER MOVES

Poll the Class

Have students answer which representation they think is best, outcome tables or tree diagrams. This can lead to the discussion of strengths and weaknesses of each representation before students fill in the table. Highlight when it is useful to use each representation, referencing the outcome table above compared to the tree diagram for rolling a die twice in the previous lesson to bring attention to size.

QUESTIONING

Consider having students write a summary of when it would be most useful to use each representation.

- When your outcome table is complete, find another pair of students to check your work with. Make any necessary changes.
- Complete the table.

Experiment	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Total Number of Outcomes in the Sample Space
Rolling a fair die two times	6	6	36

- How many outcomes are in Event A = rolling the same number?
6
- Work with your partner to complete the table showing the strengths and weaknesses/limitations of tree diagrams and outcome tables.

	Outcome Table	Tree Diagram
Strengths	<i>Answers will vary.</i> - Organizes data - Useful for large amount of outcomes	<i>Answers will vary.</i> - Organizes data - Can be used for an unlimited amount of steps
Weaknesses/ Limitations	Only uses two steps	Can get very large

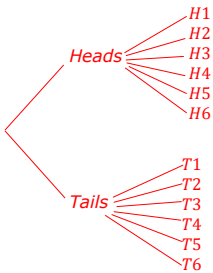


14.2.3 Guided Practice: Determining the Number of Outcomes in the Sample Space

1. A fair coin is tossed, and a fair six-sided die is rolled. Determine the sample space using either a tree diagram or an outcome table.

Answers will vary.

	Heads	Tails
1	H1	T1
2	H2	T2
3	H3	T3
4	H4	T4
5	H5	T5
6	H6	T6



2. Complete the table.

Experiment	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Total Number of Outcomes in the Sample Space
Tossing a fair coin and rolling a fair die	2	6	12

3. Work with your partner to summarize the relationship between the number of outcomes in each step of a repeated experiment and the total number of outcomes in the experiment.

Answers will vary. You can multiply the number of outcomes in each step by each other to get the total amount of outcomes.

14.2.3 GUIDED PRACTICE: DETERMINING THE NUMBER OF OUTCOMES IN THE SAMPLE SPACE

TEACHER MOVES

For a more in-depth approach for collaboration for this activity, consider having each student find another student that determined the sample space using a different representation than they did. Have students compare their representations with the following guiding questions:

- Do you have the same number of total outcomes?
- Are all outcomes included in both representations?
- What would you have done differently if using the other representation?
- How would it affect the number of outcomes in the sample space if you changed the order of the steps in the experiment?

DIFFERENTIATE INSTRUCTION

Provide sentence stems for students that may be struggling with making the connection.

- In the first step of the repeated experiment, _____
- In the second step of the repeated experiment, _____
- To get the total number of outcomes in the sample space, _____

14.2.4 BRING IT TOGETHER: DETERMINING THE SAMPLE SPACE

TEACHER MOVES

Before completing this component, refer back to question 6 in the Collaboration, question 2 in the Guided Practice, and question 1 in the Bring It Together. Consider prompting students to discuss the patterns they notice in the previous examples. Then, with their partner, develop an “I wonder” statement regarding the total number of outcomes in a sample space.

TEACHER MOVES

MA.K12.MTR.5.1: Patterns and Structure

Students use the repeated experiments explored throughout this lesson to look for similarities among the questions. Positioning students to use patterns and structure to help understand and connect different models of sample space size will help the students to develop generalizations that can be applied to other questions. Understanding these relationships will also help students better understand how to determine the theoretical probability of repeated experiments in upcoming lessons.

QUESTIONING

Using a spinner, die, and/or coin, challenge students to create a repeated experiment in which the sample space is a designated size. For example, “Create an experiment in which the sample space has 24 outcomes.”



14.2.4 Bring It Together: Determining the Sample Space

- Fill in the blue boxes with the same operation symbol to make all of the number sentences true.

Experiments	Number of Outcomes in the First Step		Number of Outcomes in the Second Step		Total Number of Outcomes in the Sample Space
Flip two fair coins	2	x	2	=	4
Roll two fair six-sided dice	6	x	6	=	36
Flip a fair coin and roll a fair die	2	x	6	=	12

- To determine the sample space for a multi-step experiment, it can be helpful to know how many total outcomes should be in the sample space. To figure this out, ☐ add ☒ multiply the number of outcomes in each step.

- For example, if an experiment consists of flipping a fair coin and spinning a spinner with eight different outcomes, the total number of outcomes in the sample space can be found by ☐ multiplying ☒ adding the number of outcomes in each step of the repeated experiment. In this case, ☐ $2 \cdot 8 = 16$. ☒ $2 + 8 = 10$.
- If a fair coin is tossed six times, the total number of outcomes in the sample space of this repeated experiment is ☐ $6 \cdot 2 = 12$. ☒ $2^6 = 64$.



14.2.5 Your Turn!

Marianne and three friends are playing a game that requires them to spin the spinner shown, then roll a fair six-sided die on each person's turn to play.



1. How many outcomes are in the sample space when...
 - a. ...the spinner is spun once?
3
 - b. ...a fair six-sided die is rolled once?
6
 - c. ...the spinner is spun, then a six-sided die is rolled?
18

14.2.5 YOUR TURN!

TEACHER MOVES

Monitor for students who skip this question until they have determined the sample space below. As one of the learning targets for this lesson, students should be able to determine the total number of outcomes in the sample space without determining the sample space itself. For students struggling with this, refer them back to question 6 in 14.2.2 Collaboration, question 2 in 14.2.3 Guided Practice, and question 1 in 14.2.4 Bring It Together. Ask students how these tables relate to the experiment of spinning this spinner and rolling a die.

14.2.5 YOUR TURN! (CONTINUED)

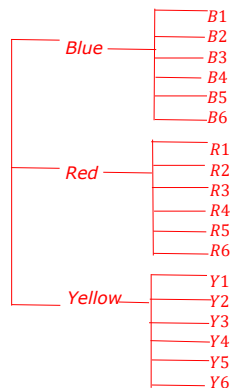
DIFFERENTIATE INSTRUCTION

Students who struggle with building the representations may benefit from using graph paper and/or rulers to create outcome tables. As an alternative, students can use software such as Microsoft Excel and Word to create the outcome tables and tree diagrams. For visual learners, provide an option to create tree diagrams with pictures instead of words and numbers.

2. Use a tree diagram or outcome table to determine the sample space.

Answers will vary.

	Blue	Red	Yellow
1	B1	R1	Y1
2	B2	R2	Y2
3	B3	R3	Y3
4	B4	R4	Y4
5	B5	R5	Y5
6	B6	R6	Y6



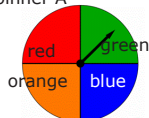
3. Write out the sample space for one player's turn.
{B1, B2, B3, B4, B5, B6, R1, R2, R3, R4, R5, R6, Y1, Y2, Y3, Y4, Y5, Y6}



14.2.6 Wrap-Up: Ticket Out the Door

Spinner A is shown.

Spinner A



1. Determine the number of outcomes in the sample space when Spinner A is spun twice.
16
2. Explain which method you would choose to determine the sample space for this random process: a list, table, or tree diagram.
Answers will vary.
3. Write the sample space for this repeated experiment.
 $\{RG, RO, RB, RR, GR, GO, GB, GG, OR, OG, OB, OO, BG, BO, BR, BB\}$

14.2.6 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

If students are struggling to identify the sample space, consider having them not only explain which representation they would use, but also have them build the tree diagram or outcome table. Not only will this help them to list the sample space, but it also serves as another demonstration of knowledge on the lesson.